MAS508-G Control Theory
Lecture 2: System Modelling

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Lecture 2c: Linearization

Book “Control Systems Engineering”, Chapter 2.11

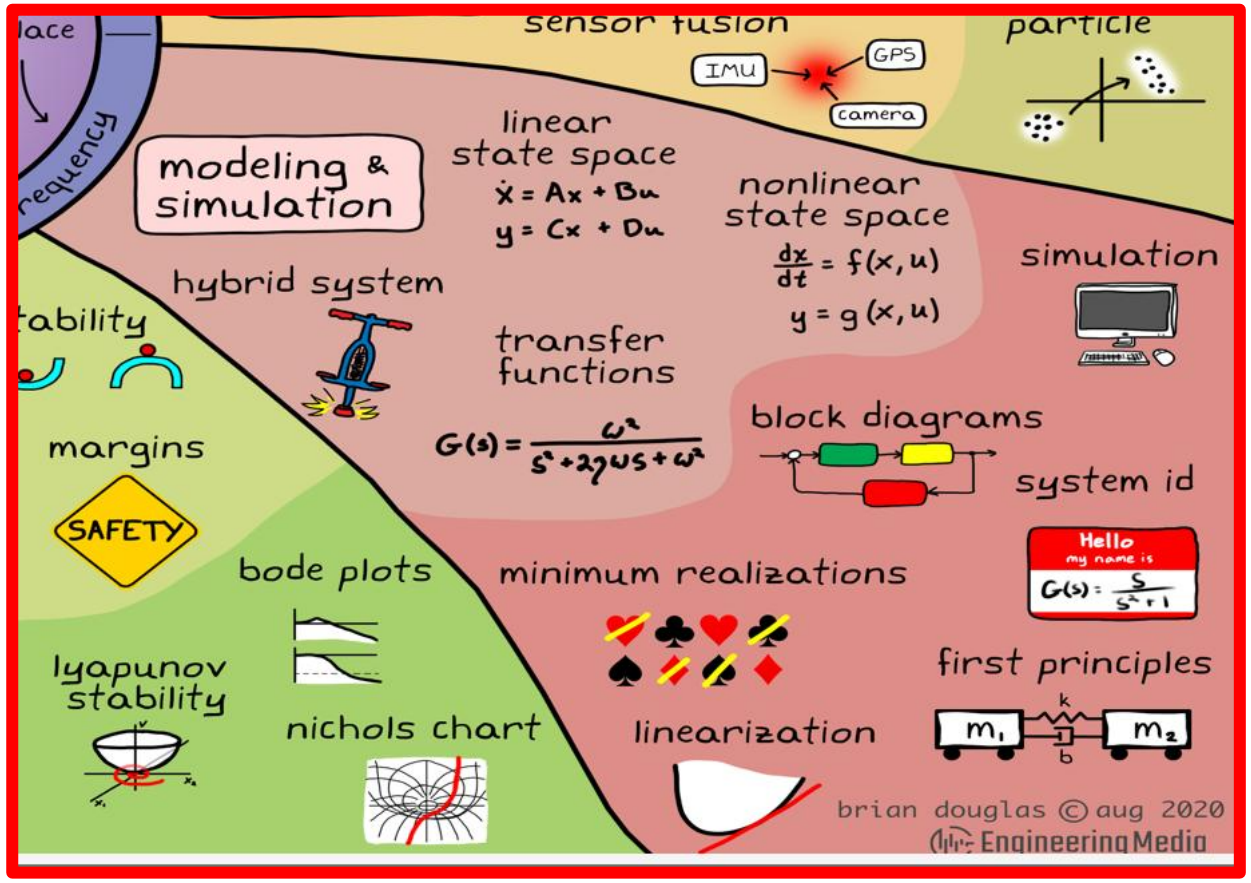
Lecture 2c Linearization

a) Review of Theoretical Modeling
   - Differential equations
   - Transfer functions
   - Block diagrams
   - Frequency response

b) State-Space Representation

c) Linearization

d) Experimental Modeling



State-Space Representation

General form:

$$\dot{x} = f(x, u)$$
$$y = h(x, u)$$

Where:

x = vector of state variables $(n \times 1)$

u = the control input vector $(r \times 1)$

y = is the output vector $(m \times 1)$

This is for a **n^{th} order system with r inputs and m outputs.**

This can describe both *linear* and *nonlinear* systems.

State-Space: Linear system

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Where:

A = system matrix $(n \times n)$

B = input matrix $(n \times r)$

C = output matrix $(m \times n)$

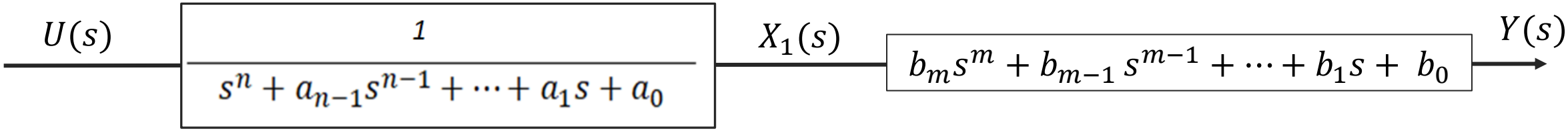
D = feedforward matrix $(m \times r)$

The system state

$$x = [x_1 \quad x_2 \quad x_3 \ldots x_{n-1} \quad x_n]^T$$

Convert TF to SS: general form

$$y^{(n)} + a_{n-1}\, y^{(n-1)} + \cdots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + b_{m-1}\, u^{(m-1)} + \cdots + b_1 \dot{u} + b_0 u$$
$$(s^n + a_{n-1}\, s^{n-1} + \cdots + a_1 s + a_0) Y(s) = (b_m s^m + b_{m-1}\, s^{m-1} + \cdots + b_1 s + b_0) U(s)$$



$$\frac{Y(s)}{U(s)} = \frac{s^m + b_{m-1}\, s^{m-1} + \cdots + b_1 s + b_0}{s^n + a_{n-1}\, s^{n-1} + \cdots + a_1 s + a_0}$$

$$\dot{x} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [b_0 \quad b_1 \quad \ldots \quad b_{m-1} \quad b_m]\, x$$

Internal state: $x(t) = \left[x_1 \quad \dot{x}_1 \quad \ldots \quad x^{(n-1)}\right]$

Convert State-Space to Transfer Function

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

Take Laplace transform for the state space model:

$$sX(s) = AX(s) + BU(s)$$
$$Y(s) = CX(s) + DU(s)$$

Solving:

$$X(s) = (sI - A)^{-1}BU(s)$$

$$Y(s) = C(sI - A)^{-1}BU(s) + DU(s) = [C(sI - A)^{-1}B + D]U(s)$$

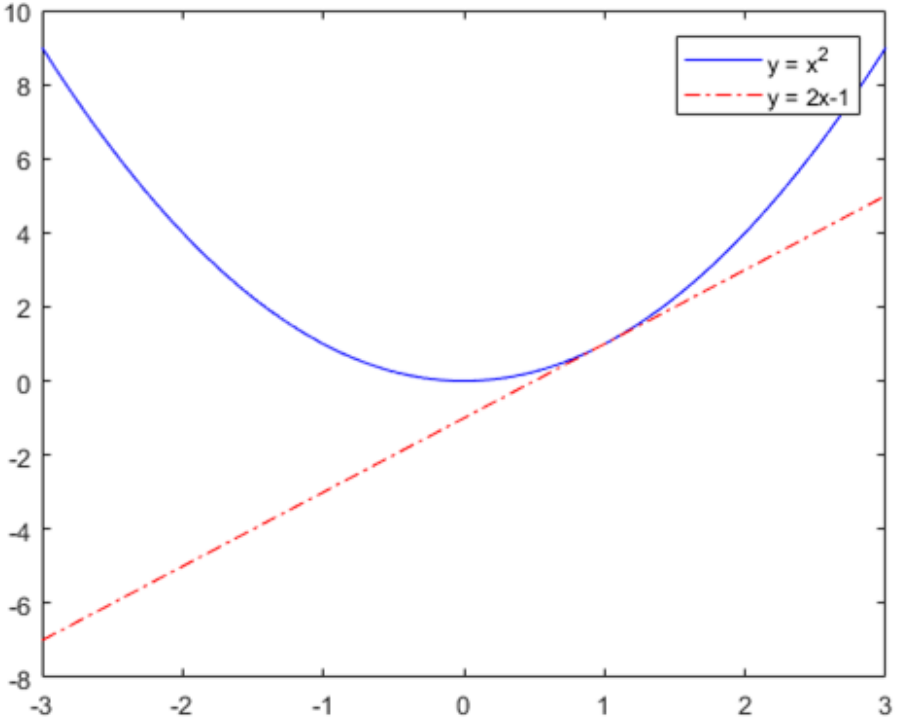
If it is SISO system $\boxed{T(s) = C(sI - A)^{-1}B + D}$

What is linearization?

Process of approximating a nonlinear function with a linear-based one.

A nonlinear function $y = x^2$ is linearized at the operating point $x_0 = 1$.

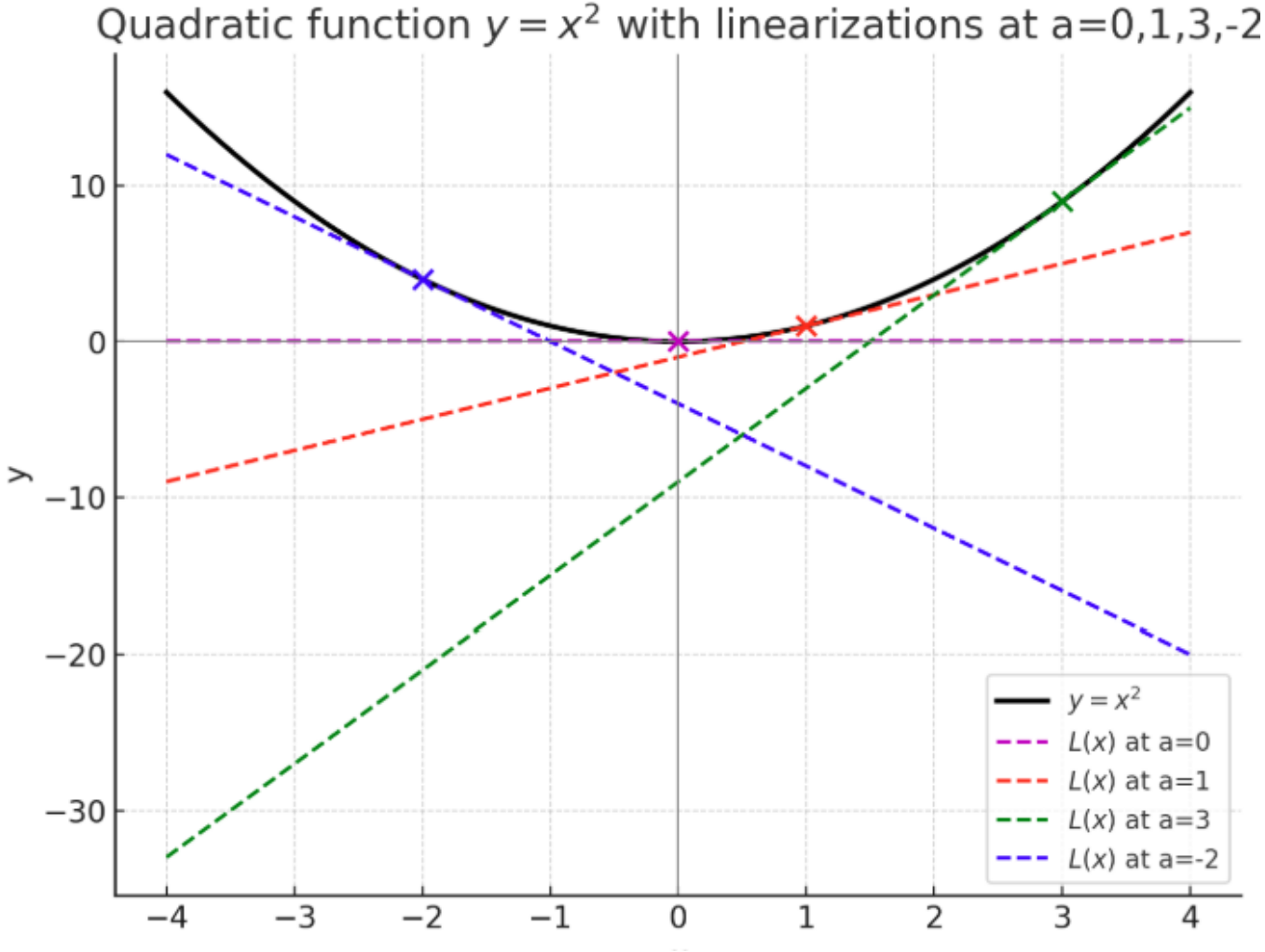
$$y = 2x - 1$$



What is linearization?

A nonlinear function $y = x^2$ is linearized at the operating point $x_0 = a$:

- $a = 0$: $L(x) = 0$
- $a = 1$: $L(x) = 2x - 1$
- $a = 3$: $L(x) = 6x - 9$
- $a = -2$: $L(x) = -4x - 4$



Why do you need linearization?

- Most systems are nonlinear
- Nonlinear models of systems can be difficult to work with
- Many systems can be represented accurately linear models
- Easier to work with a linear model (e.g. analysis and design a linear controller)

For example, Segway is a nonlinear system where the balance controller could be designed from the linearized model.



Linearization

- Linearization creates a **linear approximation** of a nonlinear system that is valid about the linearization, or operating point (equilibrium point)

- Equation to linearize a nonlinear function $f(x)$ with one variable at an operating point x_0.

$$f_{lin}(x) = f(x_0) + \left.\frac{df(x)}{dx}\right|_{x=x_0} (x - x_0)$$

(Using Taylor series expansion)

Any smooth function, $f(x)$, can be expressed as a polynomial in the neighborhood of a point a. Assuming all derivatives of the function exist at a.

$$f(x) = f(a) + \frac{(x-a)}{1!} f^{(1)}(a) + \frac{(x-a)^2}{2!} f^{(2)}(a) + \cdots + \frac{(x-a)^n}{n!} f^{(n)}(a) + \cdots$$

Linearizing Two-Variable Equation

- Equation to linearize a nonlinear function $f(x, y)$ with two variables at the operating point (x_0, y_0).

$$f_{lin}(x,y) = f(x_0,y_0) + \left.\frac{\partial f(x,y)}{\partial x}\right]_{\substack{x=x_0\\y=y_0}} (x - x_0) + \left.\frac{\partial f(x,y)}{\partial y}\right]_{\substack{x=x_0\\y=y_0}} (y - y_0)$$

General Linearization Formula

- Equation to linearize a multi-variable nonlinear function $f(x)$ at the operating point x_0, where $x = (x_1, x_2, \dots, x_n)$, and $x_0 = (x_{10}, x_{20}, \dots, x_{n0})$.

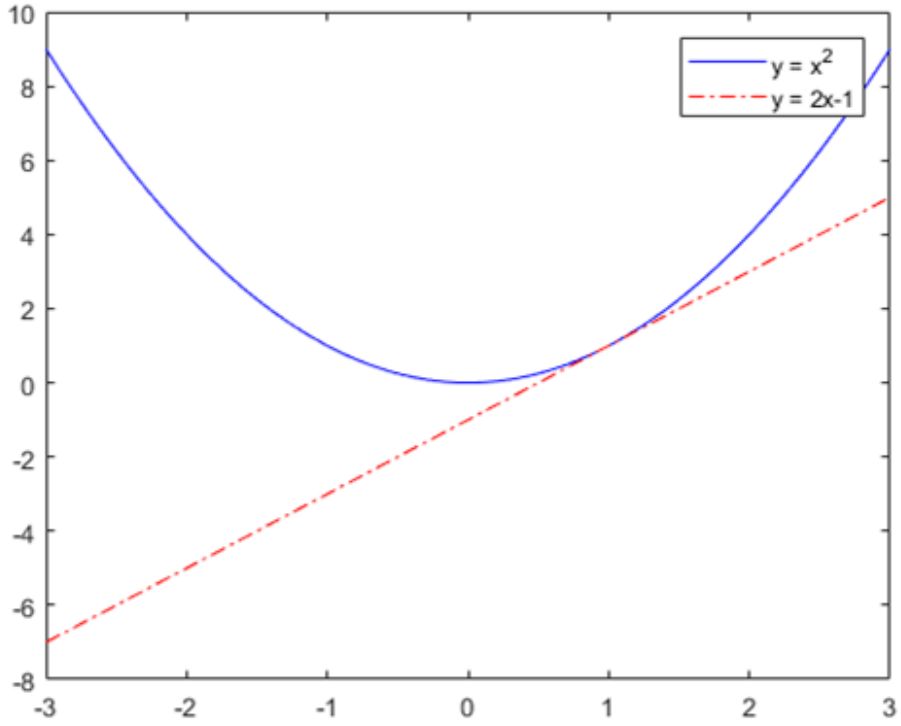
$$\mathrm{f_{lin}}(\mathbf{x}) = \mathrm{f}(\mathbf{x_0}) + \left.\frac{\partial f(\boldsymbol{x})}{\partial x_1}\right]_{x=x_0} (x_1 - x_{10}) + \left.\frac{\partial f(\boldsymbol{x})}{\partial x_2}\right]_{x=x_0} (x_2 - x_{20}) + \cdots + \left.\frac{\partial f(\boldsymbol{x})}{\partial x_n}\right]_{x=x_0} (x_n - x_{n0})$$

Example 1: Function with square term

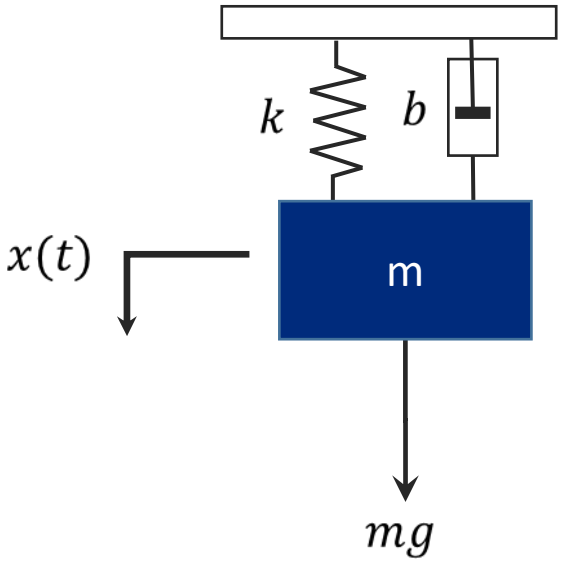
Problem: Linearize a nonlinear function $y = x^2$ at the operating point $x_0 = 1$.

$$f(x) = x^2$$
$$\frac{df(x)}{dx} = 2x$$

$$f_{lin}(x)$$
$$= f(x_0) + \left.\frac{df(x)}{dx}\right|_{x=x_0} (x - x_0)$$
$$= x_0^2 + 2x_0(x - x_0)$$
$$= 1 + 2(x - 1)$$
$$= 2x - 1$$



Example 2: Nonlinear Spring Damper



Nonlinear equation of motion for this system is: $m\ddot{x} + b\dot{x} + kx^2 = mg$

Nonlinear spring

The point of linearization $x := x_0$, or operating point, is where the mass is a rest $(\ddot{x} = 0, \dot{x} = 0)$:

$$kx_0^2 = mg$$

$$x_0 = \sqrt{\frac{mg}{k}}$$ operating point

Let $f(x) := kx^2$ and apply the linearization equation:

$$f_{lin}(x) = f(x_0) + \left.\frac{\partial f(x)}{\partial x}\right]_{x=x_0} (x - x_0)$$

$$f_{lin}(x) = kx_0^2 + 2kx]_{x=x_0}(x - x_0) = kx_0^2 + 2kx_0\,(x - x_0) = \boldsymbol{mg + 2kx_0\,(x - x_0)}$$

Linear approximation of spring force

Example 2: Simplifying

Nonlinear EOM: $m\ddot{x} = mg - b\dot{x} - kx^2$

$$x_0 = \sqrt{\frac{mg}{k}}$$

Linear EOM : $m\ddot{x} = mg - b\dot{x} - (mg + 2kx_0(x - x_0))$

$$m\,\ddot{x} = -b\dot{x} - 2kx_0(x - x_0)$$

$$\boldsymbol{m\,\ddot{x} = -b\dot{x} - 2\sqrt{kmg}\,(x - \sqrt{mg/k})}$$

Put EOM w.r.t. to operating point x_0. Define a coordinate transformation as $\boldsymbol{x = \delta x + x_0}$

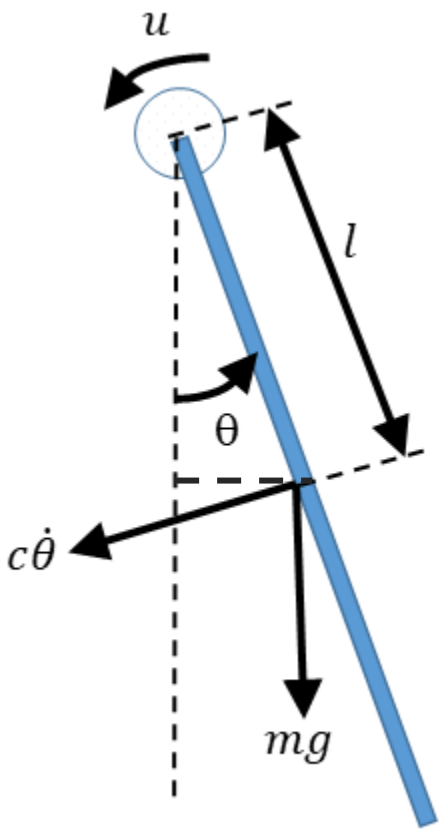
to get: $$m\frac{d^2(\delta x + x_0)}{dt^2} = -b\frac{d(\delta x + x_0)}{dt} - 2\sqrt{kmg}(\delta x)$$

Given mass is at rest at operating point: $\dot{x}_0 = 0, \ddot{x}_0 = 0$

Simplifying equation gives: $$\boldsymbol{m\frac{d^2(\delta x)}{dt^2} + b\frac{d(\delta x)}{dt} + 2\sqrt{kmg}\,\delta x = 0}$$

Equation describing the motion of the mass about the operating point.

Example 3: Pendulum



Nonlinear term in equation

Nonlinear equation of motion for this system is:

$$ml^2\ddot{\theta} + c\dot{\theta} + mgl\sin\theta = u$$

Where: *m* is the mass of the pendulum, *c* is the viscous damping, *g* is the gravitational constant, *l* is the distance to the center of mass, and *u* is the applied torque at the pivot.

Linearize the nonlinear function $\sin(\theta)$ at operating point $\theta := \theta_0 = 0$ and $u := u_0 = 0$:

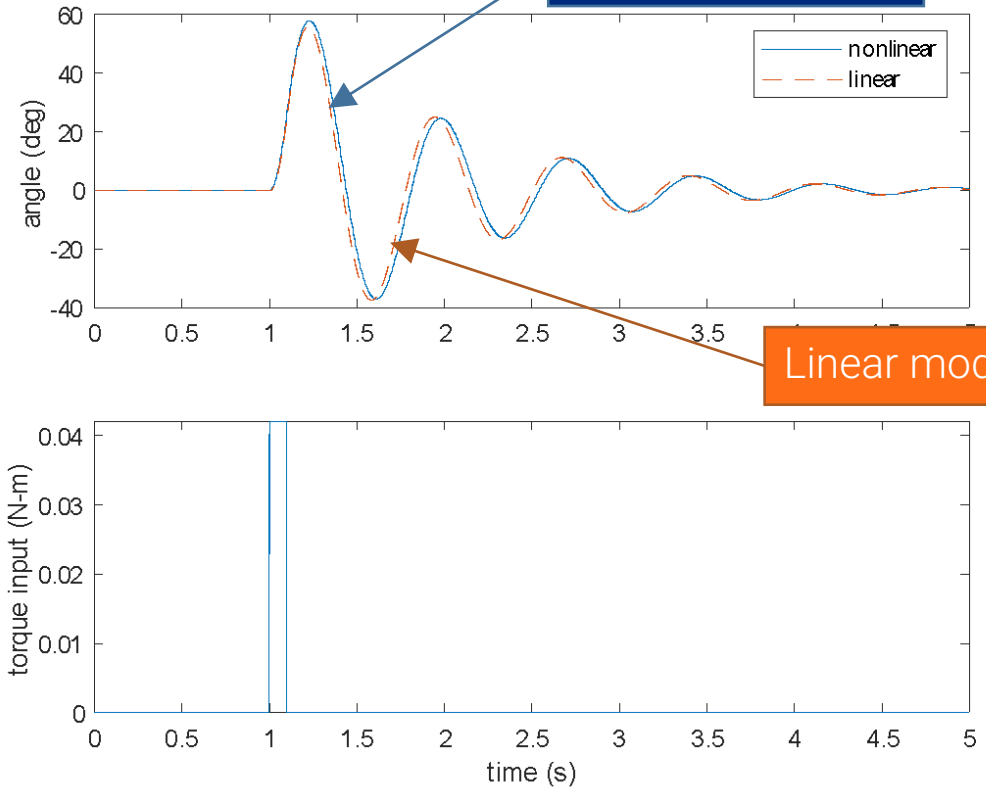
$$\boldsymbol{sin(\theta) \approx sin(\theta_0) + cos(\theta_0)(\theta - \theta_0) = \theta}$$

The linearization equation is

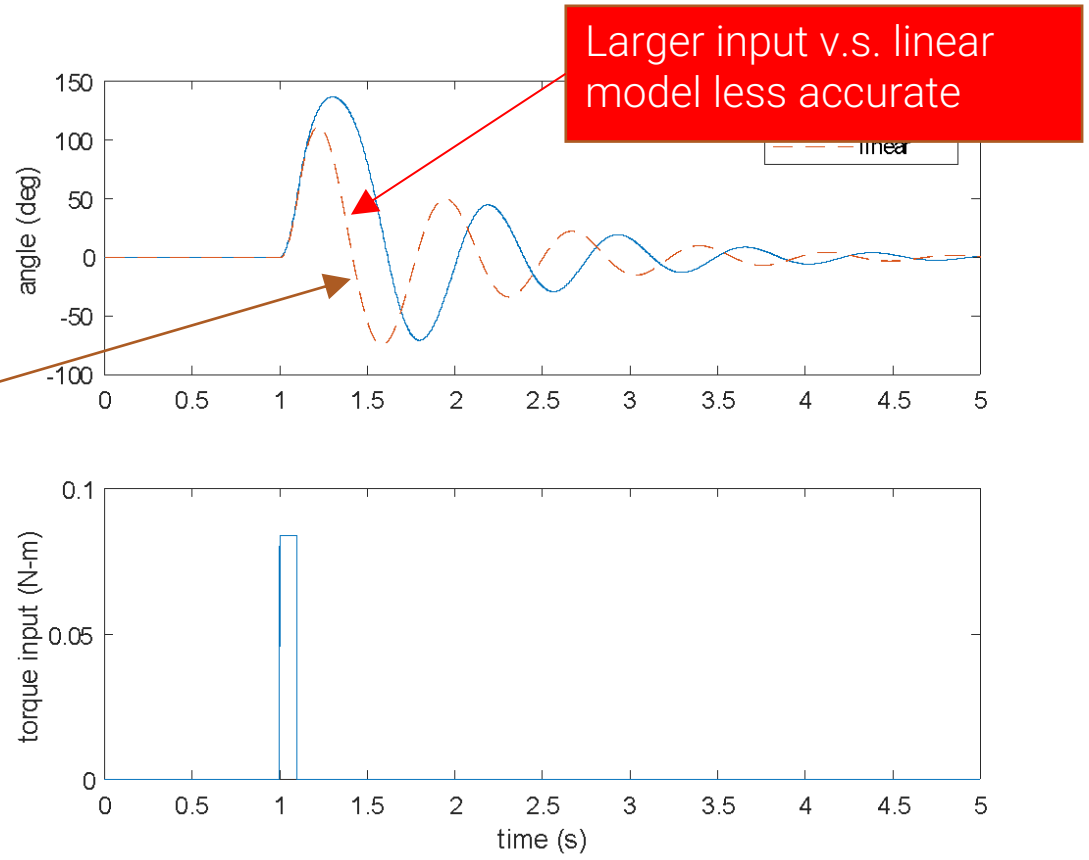
$$\boldsymbol{ml^2\,\ddot{\theta} + c\,\dot{\theta} + mgl\,\theta = u}$$

Comparison: Linear vs. Nonlinear

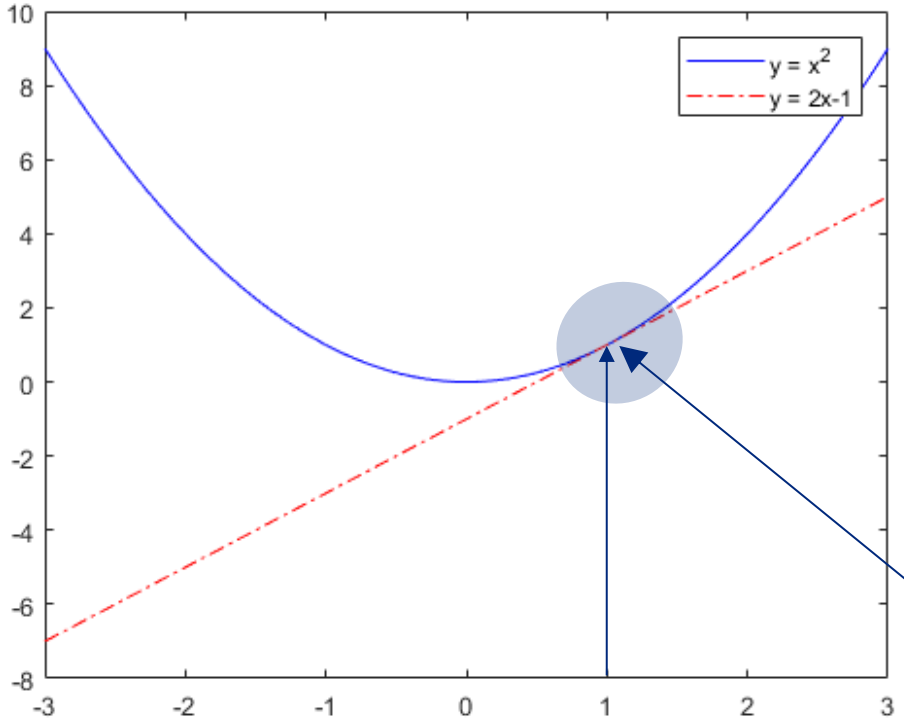
Lower torque input



Higher torque input

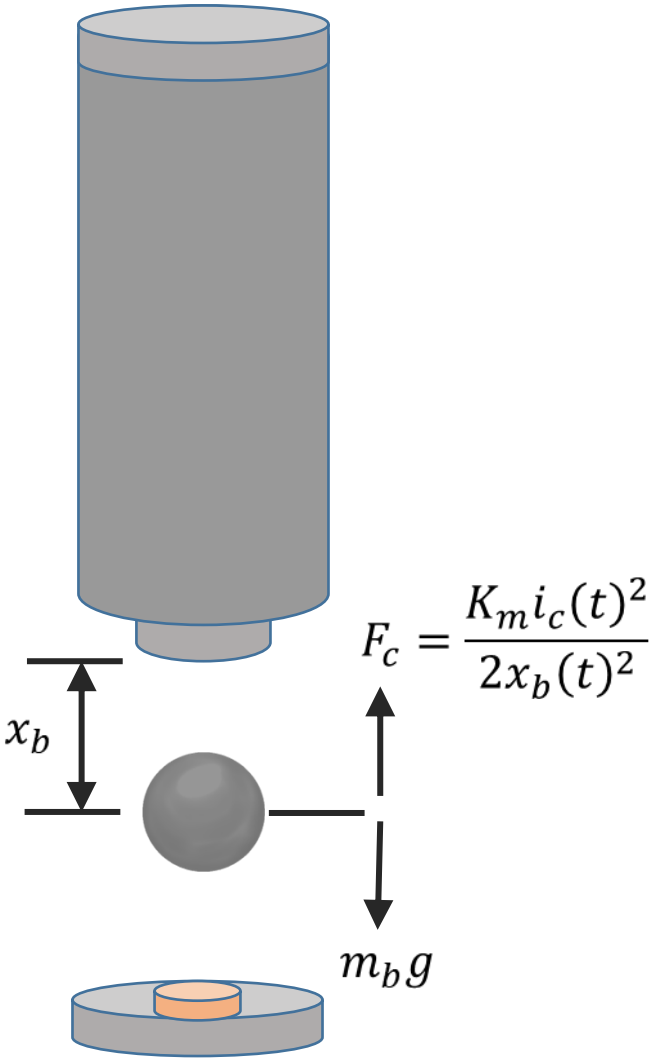


Operating Region



Linearized function is only valid for a certain region about the operating point.

Example 4: Magnetic Levitation



Nonlinear EOMs: $m_b \ddot{x}_b(t) = m_b g - F_c$

where F_c is the attractive force exerted on the ball from the magnet defined as

$$m_b \ddot{x}_b(t) = m_b g - \frac{K_m i_c(t)^2}{2x_b^2(t)}$$

Electromagnet force is nonlinear

Where K_m is the electromagnetic force constant of the electromagnet.

The operating point (x_{b0}, i_{c0}) is defined *as the point where the ball is levitating motionless against gravity*:

$$\mathbf{0} = \boldsymbol{m_b g} - \frac{\boldsymbol{K_m i_{c0}^2}}{\boldsymbol{2x_{b0}^2}}$$

$$m_b g = \frac{K_m i_{c0}^2}{2x_{b0}^2} \quad \text{or} \quad K_m = \frac{2x_{b0}^2 m_b g}{i_{c0}^2}$$

Linearizing Equation

Nonlinear EOMs: $f(x_b, i_c) \coloneqq m_b \ddot{x}_b(t) = m_b g - \frac{K_m i_c(t)^2}{2x_b^2(t)}$

Linearize this in terms of the operating point (x_{b0}, i_{c0}):

$$\mathrm{f_{lin}(x,y)} = \mathrm{f(x_{b0}, i_{c0})} + \left.\frac{\partial f(x_b, i_c)}{\partial x_b}\right]_{\substack{x_b = x_{b0} \\ i_c = i_{c0}}} (x_b(t) - x_{b0}) + \left.\frac{\partial f(x_b, i_c)}{\partial i_c}\right]_{\substack{x_b = x_{b0} \\ i_c = i_{c0}}} (i_c(t) - i_{c0})$$

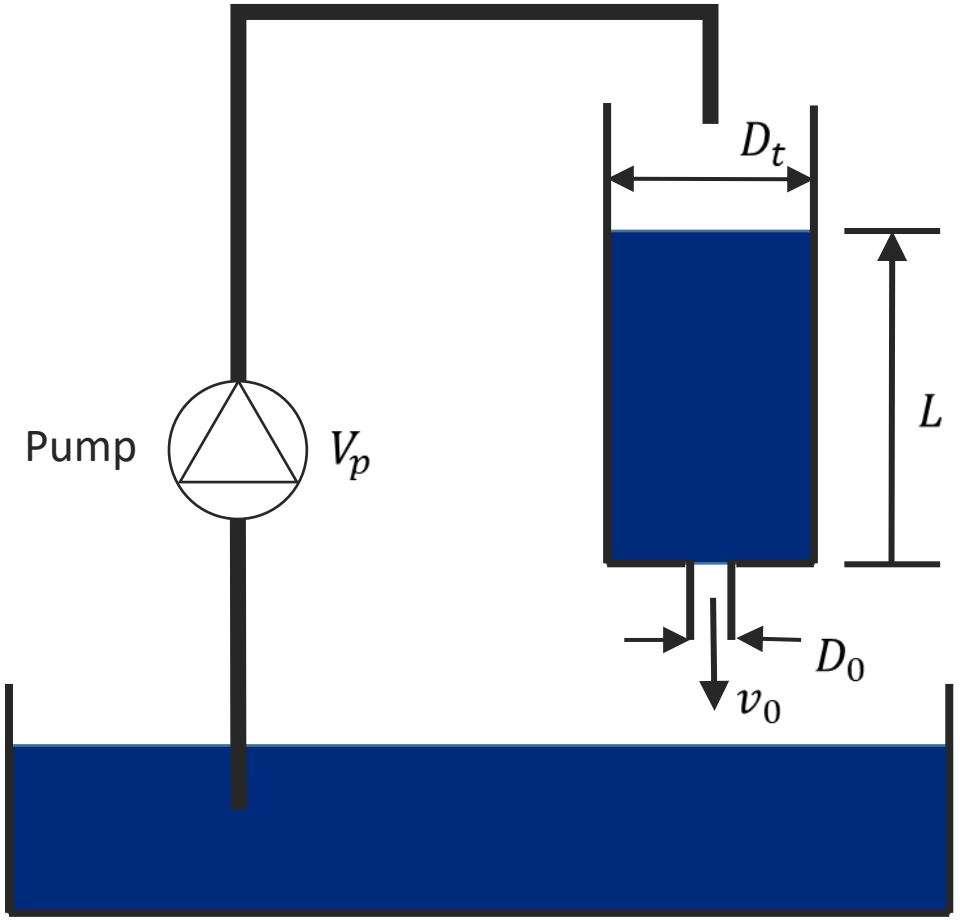
$$\mathrm{f_{lin}}(x_b, i_c) = \underbrace{m_b g - \frac{K_m i_{c0}^2}{x_{b0}^2}}_{=0} + \frac{K_m i_{c0}^2}{x_{b0}^3}(x_b(t) - x_{b0}) - \frac{K_m i_{c0}}{x_{b0}^2}(i_c(t) - i_{c0})$$

Substituting the expression $K_m = \frac{2x_{b0}^2 m_b g}{i_{c0}^2}$, $\mathrm{f_{lin}}(x_b, i_c) = \frac{2m_b g}{x_{b0}}(x_b(t) - x_{b0}) - \frac{2m_b g}{i_{c0}}(i_c(t) - i_{c0})$

and the air gab and current w.r.t. equilibrium point: $\delta x_b = x_b - x_{b0}$ and $\delta i_c = i_c - i_{c0}$

$$\boxed{\frac{d^2 \delta x_b(t)}{dt^2} = \frac{2g}{x_{b0}} \delta x_b(t) - \frac{2g}{i_{c0}} \delta i_c(t)}$$

Water Tank Level



Nonlinear EOM of water level w.r.t. to pump voltage:

$$A_t \frac{dL(t)}{dt} = K_p V_p(t) - A_o\sqrt{2gL(t)}$$

Outflow is nonlinear

Where:

- $L(t)$ is the level inside the tank,
- A_t is area in the tank,
- Ao is the area of the outflow valve,
- V_p is the pump voltage input, and
- K_p is the pump voltage to inflow rate constant.

Linearizing the Equation

Nonlinear EOM of water level w.r.t. to pump voltage:

$$A_t \frac{dL(t)}{dt} = K_p V_p(t) - A_o \sqrt{2gL(t)}$$

The nominal operating point, or equilibrium point, (L_0, V_{p0}) is defined as the point where *the water level is held at a constant position due to a constant inflow rate* generated by V_{p0}:

$$0 = K_p V_{p0} - A_o \sqrt{2gL_0} \quad \Rightarrow \quad V_{p0} = \frac{A_o}{K_p} \sqrt{2gL_0}$$

Linearizing the Equation

Nonlinear EOM: $$A_t \frac{dL(t)}{dt} = \left(K_p V_p(t) - A_0\sqrt{2gL(t)}\right)$$

$$f(L, V_p) = \left(K_p V_p(t) - A_0\sqrt{2gL(t)}\right), \quad \frac{\partial f(L,V_p)}{\partial L} = -A_0\sqrt{\frac{g}{2L_0}}, \quad \frac{\partial f(L,V_p)}{\partial V_p} = K_p$$

Linearize the system about the operating point (L_0, V_{p0}):

$$A_t \frac{dL(t)}{dt} \approx f(L_0, V_{p0}) + \left.\frac{\partial f(L,V_p)}{\partial L}\right]_{\substack{L=L_0 \\ V_p=V_0}} (L(t) - L_0) + \left.\frac{\partial f(L,V_p)}{\partial V_p}\right]_{\substack{L=L_0 \\ V_p=V_0}} (V_p(t) - V_{p0})$$

$$A_t \frac{dL(t)}{dt} = \left(\underbrace{K_p V_{p0} - A_o\sqrt{2gL_0}}_{=0} - A_0\sqrt{\frac{g}{2L_0}}(L(t) - L_0) + K_p(V_p(t) - V_{p0}) \right)$$

$$A_t \frac{dL(t)}{dt} = -A_0\sqrt{\frac{g}{2L_0}}(L(t) - L_0) + K_p(V_p(t) - V_{p0})$$

Simplifying the Equation

Linear equation: $$A_t \frac{dL(t)}{dt} = -A_0 \sqrt{\frac{g}{2L_0}} (L(t) - L_0) + K_p(V_p(t) - V_{p0})$$

$$A_t \frac{d\delta L(t)}{dt} = -A_0 \sqrt{\frac{g}{2L_0}}\ \delta L(t) + K_p\ \ \delta V_p(t)$$

where δL and δV_p is the level and pump voltage *relative to the equilibrium point*:

$$\delta L(t) = L(t) - L_0 \qquad \delta V_p = V_p(t) - V_{p0}$$

Transfer function: $$G(s) = \frac{\Delta L(s)}{\Delta V_p(s)} = \frac{K_p}{A_t s + A_0 \sqrt{\frac{g}{2L_0}}}$$

Ball and Beam System

- The Ball and Beam device is a classic control problem of stabilizing the ball to a desired position along the beam. The free-body diagram of a Ball and Beam system that uses a rotary servo to control the angle of the beam is illustrated in Figure .
- Newtonian Mechanics and Lagrange Methods
- The nonlinear equations of motions of the system are

$$\left(\frac{J_b}{r^2} + m\right)\ddot{x} + mg\ sin\ \alpha\ - m\ x\ \dot{\alpha}^2 = 0$$

$$(J + J_b + mx^2)\ddot{\alpha} + mg\ x\ cos\ \alpha + 2m\ x\ \dot{x}\ \dot{\alpha} = \tau$$

- Linearize the system for small angles about the beam horizontal, $\alpha_0 = 0$, and when ball is stabilized in the center of the beam, $x_0 = 0$, and not moving, $\dot{\theta}_0 = 0$.

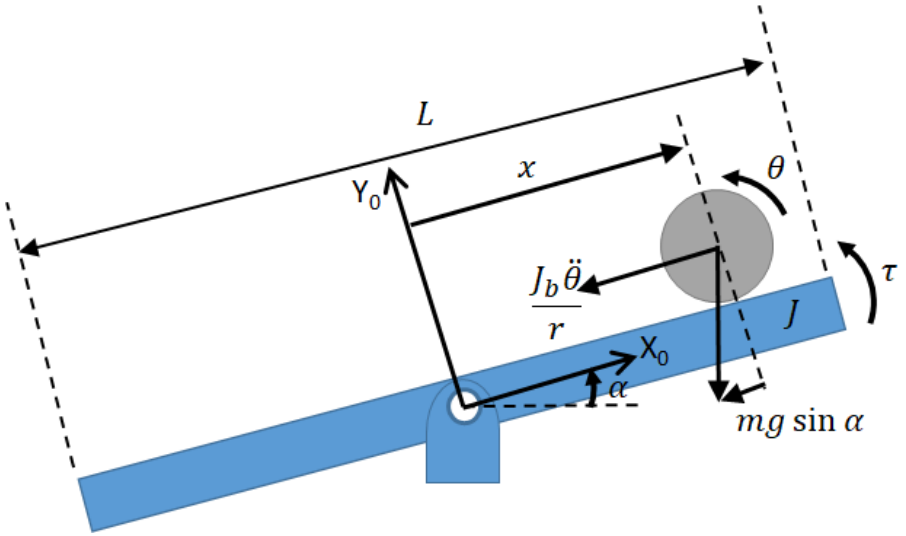


Figure - Free body diagram of Ball and Beam system

where *m and* J_b is the mass and moment inertia of the ball, *r* is the radius of the ball, *x* is the linear displacement of the ball along the beam, τ is the applied torque about the beam (the control variable), *L* is the total beam length, α is the angle of the beam relative to the horizontal, and *J* is the moment of inertia of the beam about its rotating axis.

Ball and Beam System: Linearization

- At this point, the ball is balanced at $x_0 = 0$, the beam angle is horizontal, $\alpha_0 = 0$, the ball's and beam's velocity and acceleration are zero, $\dot{x}_0 = 0, \ddot{x}_0 = 0, \ddot{\alpha}_0 = 0, \dot{\alpha}_0 = 0$. Therefore, the torque at the operating point is effectively zero,

$$\tau_0 = (J + J_b + mx_0^2)\ddot{\alpha}_0 + mg\, x_0\, cos\, \alpha_0 + 2m\, x_0\, \dot{x}_0\, \dot{\alpha}_0 = 0$$

- Assigning

$$f_1(\ddot{x}, \alpha, x, \dot{\alpha}) = \left(\frac{J_b}{r^2} + m\right)\ddot{x} + mg\, sin\, \alpha\ - m\, x\, \dot{\alpha}^2$$

$$f_2(x, \alpha, \ddot{\alpha}, \tau) = (J + J_b + mx^2)\ddot{\alpha} + mg\, x\, cos\, \alpha + 2m\, x\, \dot{x}\, \dot{\alpha} - \tau$$

- Linearizing the first equation

$$f_{1,lin}(\ddot{x}, \alpha, x, \dot{\alpha}) = f_{1,lin}(\ddot{x}_0, \alpha_0, x_0, \dot{\alpha}_0) + \frac{\partial f_1(\ddot{x}, \alpha)}{\partial \ddot{x}}\Big]_{\ddot{x}=\ddot{x}_0\ \alpha=\alpha_0}(\ddot{x} - \ddot{x}_0) + \frac{\partial f_1(\ddot{x}, \alpha)}{\partial \alpha}\Big]_{\ddot{x}=\ddot{x}_0\ \alpha=\alpha_0}(\alpha - \alpha_0)$$
$$+ \frac{\partial f_1(\ddot{x}, \alpha)}{\partial \dot{\alpha}}\Big]_{x=x_0, \dot{\alpha}=\dot{\alpha}_0}(\dot{\alpha} - \dot{\alpha}_0) + \frac{\partial f_1(\ddot{x}, \alpha)}{\partial x}\Big]_{x=x_0\ \dot{\alpha}=\dot{\alpha}_0}(x - x_0)$$

$$\boldsymbol{f_{1,lin}(\ddot{x}, \alpha) = \left(\frac{J_b}{r^2} + m\right)(\ddot{x} - \ddot{x}_0) + mg(\alpha - \alpha_0) = \left(\frac{J_b}{r^2} + m\right)\ddot{x} + mg\ \alpha}$$

Ball and Beam System: Linearization

- Linearizing the 2nd equation based on the variable $\mathrm{x} = [x, \alpha, \ddot{\alpha}, \tau, \dot{x}\ , \dot{\alpha}]$

$$f_{2,lin,}(x)$$
$$= f_{2,lin}(x_0) + \frac{\partial f_2(x)}{\partial x}]_{\mathrm{x=x_0}}(x - x_0) + \frac{\partial f_2(x)}{\partial \alpha}]_{\mathrm{x=x_0}}(\alpha - \alpha_0) + \frac{\partial f_2(x)}{\partial \ddot{\alpha}}]_{\mathrm{x=x_0}}(\ddot{\alpha} - \ddot{\alpha}_0) + \frac{\partial f_2(x)}{\partial \tau}]_{\mathrm{x=x_0}}(\tau - \tau_0)$$
$$+ \frac{\partial \mathrm{f_2(x)}}{\partial\ \dot{\mathrm{x}}}]_{\mathrm{x=x_0}}(x - x_0) + \frac{\partial f_2(x)}{\partial \dot{\alpha}}]_{\mathrm{x=x_0}}(\dot{\alpha} - \dot{\alpha}_0)$$

- Evaluating the derivatives

$$f_{2,lin}(x) = (2mx + \boldsymbol{mgcos\ \alpha} + 2m\ \dot{x}\ \dot{\alpha})]_{\mathrm{x=x_0}}(x - x_0) + (-mgx\ sin\ \alpha\ )]_{x=x_0}(\alpha - \alpha_0)$$
$$+(\boldsymbol{J + J_b} + mx^2)]_{\mathrm{x=x_0}}(\ddot{\alpha} - \ddot{\alpha}_0) + \boldsymbol{(-1)}]_{\mathrm{x=x_0}}(\tau - \tau_0) + (2m\ x\ \dot{\alpha}\ )]_{\mathrm{x=x_0}}(x - x_0) + (2m\ x\ \dot{\alpha}\ )\ ]_{\mathrm{x=x_0}}(\dot{\alpha} - \dot{\alpha}_0)$$

- At steady-state all the initial conditions are zero.

$$f_{2,lin}(x) = mg(x - x_0) + (J + J_b)(\ddot{\alpha} - \ddot{\alpha}_0) - (\tau - \tau_0)$$
$$\boldsymbol{f_{2,lin}(x) = mgx + (J + J_b)\ddot{\alpha} - \tau}$$

Linear model: Ball and Beam System

- The linearized equation of motion is

$$\left(\frac{J_b}{r^2}+m\right)\ddot{x}+mg\ \ \alpha=0$$

$$(J+J_b)\ddot{\alpha}+mgx=\tau$$

- State-space model

$$\dot{\mathrm{x}}=\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & -\dfrac{mgr^2}{J_b+mr^2} & 0 \\ 0 & 0 & 0 & 1 \\ -\dfrac{mg}{J+J_b} & 0 & 0 & 0 \end{bmatrix}\mathrm{x}+\begin{bmatrix}0\\0\\0\\1\end{bmatrix}u$$

$$y=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0\end{bmatrix}\mathrm{x}=\begin{bmatrix}x\\ \alpha\end{bmatrix},\qquad \mathrm{x}=[x\quad \dot{x}\quad \alpha\quad \dot{\alpha}]^T,\qquad u=\tau$$

Linearization: Takeaways

Benefits

- Easier to work with linear equations vs. nonlinear ones
- Analysis of linear system is easier and often sufficient
  - Easier to design a linear control system

Drawbacks

- Model (and control) only valid about linearization point
- Using the full nonlinear model has advantages:
  - Can lead to performance improvements
  - Valid over larger operating range